A02 under inherited default memory policy

Private genome allocation

Pages initially concentrated near loading activity

Automatic NUMA balancing migration and fault work

96 mapping workers

Runtime parent

Post-A02 attribution

Serialized parser remains

94.16 equivalent mapping CPUs 0.75 worker-equivalents lock wait

A02b queue rejected insufficient scheduling headroom

External interleave screen mapping interval 34.63 s to 23.30 s

A05 accepted in blackstar.2

Auto policy eligible? alignReads + private genome at least 64 threads + multiple nodes

Yes

No

Caller supplied NUMA policy?

Retain inherited default lower threads, shared mode, or unsupported platform

Yes

Default

Preserve bind, preferred, local, or interleave subset

Interleave across process-allowed memory nodes

Load same genome run same mapping kernels

Exact 91de892 public evidence at 96 threads

Uncompressed full corpus 74.01 s to 57.35 s 22.51% median gain

Peak RSS change -0.13% 3 of 3 paired outputs passed 64-thread boundary gain 14.27%

Gzip support 78.31 s to 62.25 s 5 of 5 paired outputs passed

Gzip performance not formally accepted A02 control CV 4.28% exceeded gate